

StoryNest

An AI-Powered Bilingual (English + Hindi) Reading & Literacy Platform

Creating personalized stories, interactive narration, and gamified comprehension experiences for children (Ages 3–12).

■ **Google Gemini AI**

Generates personalized bilingual stories + vocabulary + quizzes atomically.

■ **Bilingual Literacy**

English + Hindi text side-by-side with synchronized Web Speech TTS narration.

■ **Gamified Quizzes**

Comprehension checks, star rewards, daily streaks & printable certificates.

■ **Dual Dashboards**

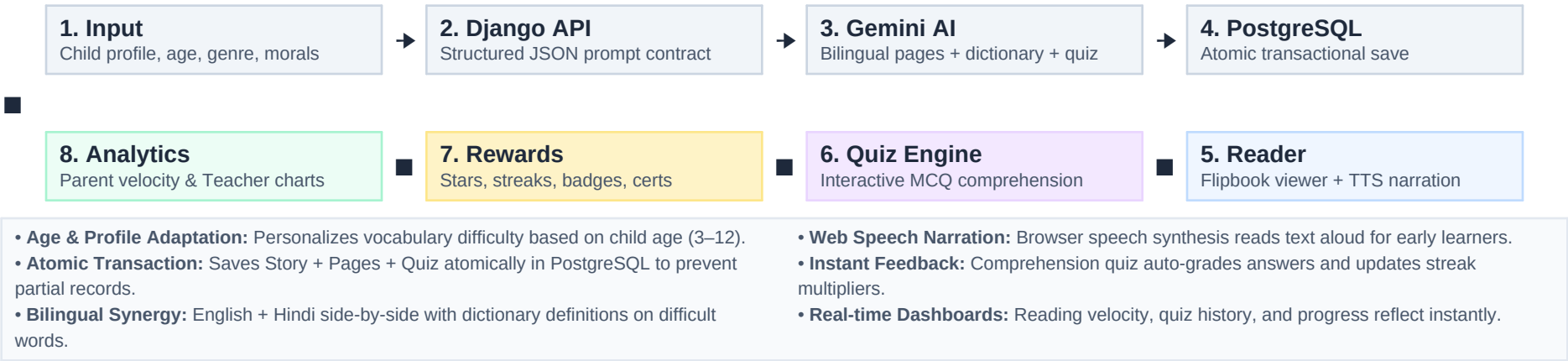
Dedicated Parent & Teacher portals with learning velocity & reading analytics.

■ **Live Explanation Guide:**

“StoryNest ek AI-powered educational platform hai jo children ke liye personalized English-Hindi stories generate karta hai aur interactive reading, quizzes aur gamification ke through literacy improve karta hai.”

Slide 2 — Core Learning & AI Architecture Flow

Visual step-by-step pipeline from child personalization to gamification and analytics



Slide 3 — User Roles, Permissions & Security Architecture

Strict role separation and row-level ownership validation across all portals

■■■■■ PARENT PORTAL • Manage child profiles & interests • Generate & read custom stories • Track daily streaks, goals & bedtime schedule • View AI vocabulary growth & progress reports • Download milestone achievement certificates	■■■ TEACHER HUB • Classroom management & unique join codes • Student rosters & individual skill breakdowns • Story homework assignments with due dates • Quiz submissions grading & pass rate metrics • Automated academic PDF report cards	■■ ADMIN SUITE • System-wide user directory & role controls • Overall platform usage analytics & story counts • System health monitoring & active logs • Content moderation & database management
■ Verified Row-Level Data Isolation: • Parent A → Child A Data: HTTP 200 OK (Authorized) • Parent A → Child B Data: HTTP 404 NOT FOUND (Access Denied) • Parent B → Child A Data: HTTP 404 NOT FOUND (Protected) ■ Live Explanation Note: <i>"Backend ownership validation filters ensure karte hain ki changing an object ID in API requests will never leak or expose another parent's children, reading logs, or private data."</i>		

Slide 4 — Technology Architecture & Cloud Deployment

Production-grade decoupled stack with high performance and security guarantees

■ FRONTEND CLIENT • React 19 + Vite 8 • Tailwind CSS 4 Glassmorphism • Axios with JWT Interceptors • Framer Motion Flipbook Reader • jsPDF & html2canvas • Hosted on: Netlify / Vercel	■■ BACKEND API • Python 3.12 / Django 5 • Django REST Framework • SimpleJWT Token Rotation • Atomic Transactions • WhiteNoise Static Assets • Hosted on: Render / AWS	■■ DATABASE & AI SERVICES • PostgreSQL Cloud Database • Google Gemini 1.5 / 2.0 API • Web Speech Audio API • Environment Key Isolation • CORS Whitelisting • Storage: Managed DB
■ JWT Authentication: Access + Refresh tokens with expiry ■ Role Authorization: Route guards on Parent, Teacher, Admin ■ Ownership Validation: Child/Story access scoped strictly to user ■ Zero Secret Exposure: Gemini & DB keys strictly in .env ■ Production CORS: Whitelisted domain origins only ■ Clean Builds: 0 Vite compile errors & 0 Django check issues		

Slide 5 — Developer Golden Rules & Project Status

Standards for maintaining stability, scalability, and seamless integration

■ PARENT MODULE STATUS: 100% COMPLETE, VERIFIED & PRODUCTION READY

1. Preserve Existing Contracts

Never break or rework working Parent & Teacher logic or existing API response formats.

3. Security Testing Mandatory

Always execute cross-user ID tampering tests (asserting HTTP 404 on unauthorized data).

2. Instant Database Migrations

Any model modification must be followed immediately by `makemigrations` and `migrate`.

4. Full-Stack Verification

Always run ``npm run build`` and ``python manage.py check`` before committing changes.

■ *Final Presentation Conclusion:*

"StoryNest is not just an AI story generator; it is a complete bilingual literacy ecosystem that combines personalized storytelling, interactive reading, assessment, gamification, and learning analytics in one secure, production-ready platform."